

TECHNICAL SKILLS

AI / Machine Learning: PyTorch, Keras, Explainable AI (XAI), Adversarial Machine Learning, Deep Learning

Programming Languages: Python, JavaScript, TypeScript, Java

Frontend Technologies: Angular, React.js, Next.js, Redux, PrimeNG, HTML5, CSS3, SCSS

Backend & APIs: Node.js, Spring Boot, RESTful APIs, SQL

DevOps & Tools: Git, CI/CD, Docker, Kubernetes, WCAG 2.2 AA

EDUCATION

- **PhD in Data Science & Engineering** August 2026 - May 2030
University of South Dakota, SD, USA
- **Masters of Science in Computer Science** January 2025 - May 2026
University of South Dakota, SD, USA CGPA: 4.00
- **Bachelor of Science in Computer Science and Engineering** 2015-2020
North South University, Bangladesh CGPA: 3.34

PEER-REVIEWED PUBLICATIONS

- Wang, L., Uddin, I. I., Santosh, K. C., Zhang, C., Qin, X., & Zhou, Y. Bridging Symmetry and Robustness: On the Role of Equivariance in Enhancing Adversarial Robustness. In the Proceedings of NeurIPS 2025 [\[Link\]](#)
- Wang, L., Uddin, I. I., Qin, X., Zhou, Y., & Santosh, K. C. (2025). Explainability-driven defense: Grad-CAM-guided model refinement against adversarial threats. In Proceedings of the AAAI 2025 Summer Symposium [\[Link\]](#)
- Wang, L., Uddin, I. I., & Santosh, K. C. (2026). Expert-Guided Explainable Few-Shot Learning with Active Sample Selection for Medical Image Analysis. IEEE Journal of Biomedical and Health Informatics [\[Link\]](#)

MANUSCRIPTS IN PREPARATION

- Explainable Category Discovery [\[Work in Progress\]](#)

EXPERIENCE

- **Graduate Research & Teaching Assistant** January 2025 – Present
University of South Dakota SD, USA
 - Design deep learning architectures that are intrinsically explainable and adversarially robust, embedding transparency and security directly into model design.
 - Build and benchmark end-to-end experimental pipelines on real-world datasets to validate trustworthy AI systems through explainability and adversarial robustness evaluation.
 - Support undergraduate students in math for machine learning, Python, and JavaScript through lab sessions, and evaluate assignments to provide constructive feedback.
 - Tech Stack: Python, PyTorch
- **Software Engineer** March 2023 - December 2024
Kaz Software Limited Dhaka, Bangladesh
 - Contributed to the design and development of IBFD's Tax Research Platform, collaborating with international teams to implement new features, refactor legacy code, resolve technical challenges and in addition worked on improving platforms accessibility in compliance with Web Content Accessibility Guidelines (WCAG) 2.2 AA standards.
 - Worked closely with back-end developers to design and develop RESTful APIs for front-end integration, leveraging a full architectural understanding of the system.
 - Worked as a Front-End Software Engineer using Angular; led junior developers, ensured end-to-end application maintenance, implemented new features, and conducted R&D for feature enhancements.
 - Tech Stack: Angular, TypeScript, JavaScript, Java

• Software Engineer

July 2022 - Feb 2023

Brain Station 23 PLC

Dhaka, Bangladesh

- Designed and developed three Angular-based client applications, including two for a financial platform.
- Collaborated with back-end engineers to design and integrate APIs for seamless front-end functionality.
- Supported multiple live projects by fixing bugs, optimizing code, and implementing new features.
- Mentored junior developers on front-end technologies and software development best practices.
- Tech Stack: Angular, .NET, Oracle, SQL

• Junior Software Engineer

June 2020 - June 2022

Square Health Limited

Dhaka, Bangladesh

- Collaborated with cross-functional software engineering teams to develop new features and resolve existing system issues.
- Led the development of key functionalities, including audio/video calling, dental module integration, medicine suggestion settings, and dynamic prescription design.
- Conducted research and feasibility analysis to identify optimal implementation strategies for upcoming features.
- Evaluated and integrated tools, libraries, APIs, and frameworks to enhance performance and ensure security in a data-intensive web environment.
- Tech Stack: JavaScript, TypeScript, Angular, Dexie.js, Moment.js

VOLUNTEERING / OPEN-SOURCE CONTRIBUTION

- **Prime NG:** Contributed to the open-source project Prime NG, a widely used UI Component Library for frontend development. Provided solutions to various issues listed on their GitHub issue board, enhancing the library's functionality and user experience.

–GitHub: Prime NG Repository

- **WirHelfen.eu:** Collaborated with WirHelfen.eu, a non-profit humanitarian platform focused on aiding individuals in need. Contributed to the design and development of a responsive and user-friendly UI to assist people in accessing support

–Website: WirHelfen.eu

PROJECTS

• Tax Research Platform [Live Link]

A subscription-based international tax research platform developed by IBFD, Netherlands.

- Implemented new features and optimized existing modules through code refactoring.
- Conducted R&D on transitioning the monolithic structure to a micro front-end architecture.
- **Tech Stack:** Angular, PrimeNG, AmCharts, Java

• Jotno [Live Link]

A telemedicine platform used by thousands of doctors in Bangladesh, developed under Square Health.

- Led front-end development for both doctor and patient portals.
- Developed key modules including audio/video calling, dynamic prescription templates, and medicine suggestion settings.
- **Tech Stack:** Angular, PrimeNG, Dexie.js

• Blogging Web Application [GitHub]

A full-featured blogging platform with essential social networking functionalities.

- Applied CQRS design pattern on the backend and DRY principles on the frontend to improve modularity and scalability.
- Implemented complete user interaction workflows and data handling pipelines.
- **Tech Stack:** React.js, .NET, MobX, TypeScript, C#

• School of Wolf E-Sports Web Application [Live Link]

A static web application showcasing achievements of School of Wolf, a top FIFA Pro Club team in Bangladesh.

- Designed and developed a responsive front-end to highlight team trophies and individual player awards.
- Maintained clean, reusable code by strictly following the DRY (Don't Repeat Yourself) principle.
- **Tech Stack:** TypeScript, Angular, SCSS

CERTIFICATIONS

- Angular Development Certification – HackerRank
- Angular Complete Guide – Udemy
- Full Stack Development with Angular and .NET – Udemy
- Full Stack Development with React.js and .NET – Udemy